

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10

Total Marks:50

Annexure A

Constraint - Based Problem-Solving

Duration: 2Hrs

1. Develop a Python-based Reinforcement Learning model using the OpenAI Gym environment to solve a Grid World navigation problem where the agent must reach the goal while avoiding obstacles. Explain how the state space, action space, reward function, and constraints are defined.
(10 Marks)
2. Implement an ϵ -greedy Multi-Armed Bandit algorithm in Python to maximize rewards under a limited budget of 500 trials. Compare the effect of different ϵ values (0.1, 0.3, and 0.5) on exploration, exploitation, and constraint satisfaction.
(10 Marks)
3. Design a Markov Decision Process (MDP) for an autonomous robot that must reach a destination while minimizing energy consumption. Define the states, actions, transition probabilities, rewards, and energy constraints.
(10 Marks)
4. Using Python and TensorFlow/Keras, implement a value iteration algorithm to determine the optimal policy for a constrained grid environment where certain states are restricted. Explain how Bellman equations are used to obtain the optimal policy.
(10 Marks)

5. A warehouse robot has a battery capacity of only 30 minutes. Design an RL framework that enables the robot to complete the maximum number of deliveries without exceeding the battery limit. Explain the constraints, policy, and reward design.

(10 Marks)

6. Implement a Reinforcement Learning agent that controls traffic signals at an intersection while ensuring that emergency vehicles receive immediate priority. Explain how exploration, exploitation, and policy learning are modified to satisfy the safety constraint.

(10 Marks)

7. Develop a Python program using Keras/TensorFlow to simulate a reinforcement learning agent for packet routing in a wireless network with limited bandwidth. Explain how bandwidth constraints influence the reward function and policy optimization.

(10 Marks)

8. Analyse an experimental comparison between random action selection and ϵ -greedy action selection in a Multi-Armed Bandit problem under a fixed time constraint. Record cumulative rewards and explain which strategy provides better performance.

(10 Marks)

9. Design a Reinforcement Learning framework for a delivery drone that must maximize successful deliveries while avoiding no-fly zones and maintaining battery limits. Explain the MDP formulation, Bellman equation, and policy learning process.

(10 Marks)

10. Implement a Reinforcement Learning solution using Python for a smart energy management system where electricity consumption must remain below a specified threshold. Explain how reward shaping, policy learning, and feasibility constraints improve decision-making.

(10 Marks)

Code of Conduct:

I Uday Kiran (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Uday Kiran
Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Grid world Navigation using open AI Gym

Problem understanding

Design a Reinforcement learning agent that moves in a grid world environment the agent starts from the initial position and reaches the goal while avoiding obstacles.

Constraint Analysis

- Grid size : 5×5
- obstacles cannot be crossed.
- Agent can move only inside the Grid.
- Goal must be reached in minimum steps.

Solution Development

state space :

- every grid cell represents one state.
- Total states = 25.

Action space :

- up
- Down
- left
- Right

Reward function ::

- Goal = +100
- obstacle = -100

Application of concepts

- The agent learns using Q-learning.

Q-learning

$$Q(s, a) = Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

- $Q(s, a)$ = current predicted value by taking action a in state s .

- α = learning rate
- R = Reward
- γ = Discount factor

Justification

- The reward function encourages the shortest safe path while avoiding obstacles.

2. ϵ -Greedy multi-Armed Bandit

Problem understanding

- Maximise reward using only 500 trials.

Constraint Analysis

- Total trials = 500
- Three epsilon values

- $\epsilon = 0.1$

- $\epsilon = 0.3$

- $\epsilon = 0.5$

Solution Development

<u>ϵ</u>	<u>Exploration</u>	<u>Exploitation</u>
		high
0.1	low	
0.3	Medium	Medium
0.5	High	low

- Average reward is calculated after every trial.

Application of concepts

Action selection

- Random number $< \epsilon \rightarrow$ explore.

- otherwise $\rightarrow$ exploit

Justification

ϵ -greedy usually provides the highest cumulative reward because exploitation is higher.

3. MDP for Autonomous Robot

Problem understanding

Robot must reach destination while saving battery.

Constraint Analysis

- Limited battery
- Avoid unnecessary movement.
- Reach destination safely.

Solution Development

states

- Robot positions

Action

- up
- Down
- Left
- Right

4. Value iteration

Problem understanding

Finding optimal path using Bellman equation.

Constraint Analysis

- some states are restricted.
- Agent cannot enter restricted cells.

Solution Development

Bellman Equation

$$v(s) = \max_a [R + \gamma \sum_{s'} p(s'|s/a) v(s')]]$$

- Repeat until value converge.

Application of concepts

- value iteration updates all states repeatedly

Justification

- Produces optimal policy while avoiding restricted states.

Transition Probability

- 0.9 correct move
- 0.1 wrong move

Rewards

- Destination = +100
- Energy cost = -2
- Collision = -100

Application of concepts

MDP Components

- s = states
- A = Actions
- P = Transition probability
- R = Reward
- γ = Discount factor

Justification

- Robot learns shortest path with minimum energy consumption.

5- Warehouse Robot

Problem understanding

- Robot has only 30-minute battery.

Constraint Analysis

- Battery $\leq$ 30 minutes
- Maximum delivers
- Return before battery ends.

Solution Development

states :

- Battery level
- Robot position

Rewards

- Delivery = +50
- Battery exhausted = -100
- idle = -5

Policy :-

choose nearest delivery.

Application of concepts

- Battery level becomes part of state representation.

Justification

- Maximize deliveries while preventing battery failure.

LSI